NQRust-HV

Memory-Safe Enterprise Hypervisor Architecture

Rust-Based Virtualization Platform with Superior Security & Performance

NQR
Nexus Quantum Rust

MEMORY SAFETY
- Zero buffer overflows
- Nouse-after-freebugs
- Compile-time guarantees
- 90% CVE reduction

COMPETITIVE ADVANTAGES

Security	Memory safe, zero corruption vulnerabilities
Performance	100ms boot vs 15s traditional hypervisors
Density	2000+ VMs vs 200-300 on legacy systems
Cost	74% lower TCO than VMware vSphere
Sovereignty	100% transparent, auditable codebase
Reliability	Rust prevents 70% of system crashes
Efficiency	50% lower memory overhead
Innovation	Cloud-native, API-first design

PERFORMANCE METRICS
- 100ms VM boottime
- 2000+VMs per host
- 195Gbps network throughput
- 7.2M storage IOPS

MANAGEMENT & ORCHESTRATIONLAYER
EnterpriseIntegration & Automation
Multi-Protocol Management APIs

Web Console
Modern UI/UX
Real-time Dashboards

RESTAPI
RESTful Services
JSON/GraphQL

Kubernetes CRDs
Cloud Native
Operator Pattern

Terraform Provider
IaC Integration
Infrastructure as Code

Ansible Modules
Automation
Configuration Mgmt

VIRTUALIZATION CONTROLPLANE
Resource Management & Orchestration
Intelligent Scheduling & Optimization

Web Console
ModernUI/UX
Real-time Dashboards

Resource Scheduler
NUMA Aware
Load Balancing

Network SDN
Virtual Networking
Micro-segmentation

Migration Engine
Live Migration
<50ms Downtime

NQRust-HV VMM CORE (MEMORY-SAFE)
Cloud Hypervisor Foundation
Compile-Time Safety Guarantees

Cloud Hypervisor
VMM Engine
Rust Implementation

Memory Management
Safe Allocation
KSM/Ballooning

Virtio Devices
Para virtualization
High Performance I/O

Security Policies
Isolation Engine
Hardware TEE

SECURITY & COMPLIANCE
Data Sovereignty

Memory Safety
Zero CVEs
Compile time Checks

Data Sovereignty
Indonesian Compliance
UUPDP/OJK Ready

PERFORMANCE & SCALE
Ultra-Fast Operations

100ms Boot
VM Provisioning
100x Faster

2000+VMs/Host
High Density
Resource Efficiency

HYPERVISOR BACKEND LAYER
Hardware Virtualization Interface
Cross-Platform Support

Linux KVM
Primary Backend
Intel VT-x/AMD-V

Microsoft MSHV
Windows Backend
Hyper-V Integration

SOVEREIGNTY BENEFITS
- 100% open source
- Indonesian compliance
- No foreign dependencies
- Local supportteam

VM LIFECYCLE MANAGEMENT

VMM Core	Cloud Hypervisor + Custom Extensions
Memory Safety	Rust Language with Ownership Model
Hardware Backend	Linux KVM + Microsoft MSHV
Device Emulation	Virtio + vhost-user Processes
Networking	Tokio Async + Custom Protocols
Storage	NVMe + Virtio-blk + Live Migration
Security	Hardware TEE + TPM2.0 + HSM
Management	RESTAPI + gRPC + WebSocket

VM LIFECYCLE MANAGEMENT

Provisioning	100ms boot time with security validation
Resource Allocation	NUMA-aware CPU and memory assignment
Live Migration	<50ms downtime with integrity checks
Scaling	Automatmatic resource adjustment and balancing
Monitoring	Real-time performance and security metrics

ENTERPRISE DEPLOYMENT

Management Nodes
Monitoring Stack
Network Fabric
Storage Nodes
Compute Hosts
Load Balancers

Architecture Layer	Primary Tech	Language	Key Feature	Performance	Security
Management APIs	REST + GraphQL	Rust/Typescript	Multi-protocols	<10ms response	TLS 1.3 + RBAC
Control Plane	Tokio Runtime	Rust	Async processing	Linear scaling	Memory isolation
VMM Core	Cloud Hypervisor	Rust	Memory safety	100ms boot	Compile-time checks
Device Emulation	Virtio + vhost	Rust	Para-virtualization	Near-native perf	Process isolation
Hardware Backend	KVM/MSHV	C + Rust FFI	Cross-platform	Hardware accel	TEE support

Memory Safety
Zero buffer overflows
Compile-time guarantees

Performance
100ms VM provisioning
2000+ Vms per host

Sovereignty
100% open source
Indonesian compliance

LEGACY HYPERVISORS

Solution	Language	NQRust-HVAdvantage
VMware ESXi	C/C++	74% cost reduction, memory safety
Hyper-V	C/C++	5x faster boot, no licensing fees
KVM/QEMU	C	Memory safety, enterprise features
Xen	C	Better performance, modern APIs
Proxnox	C++	Professional support, compliance

CLIENT ECOSYSTEM

Web Console
CLI Tools
Terraform
Kubernetes

ARCHITECTURE LEGEND
Management Flow · Control Flow · VMM Operations · VMM Operations

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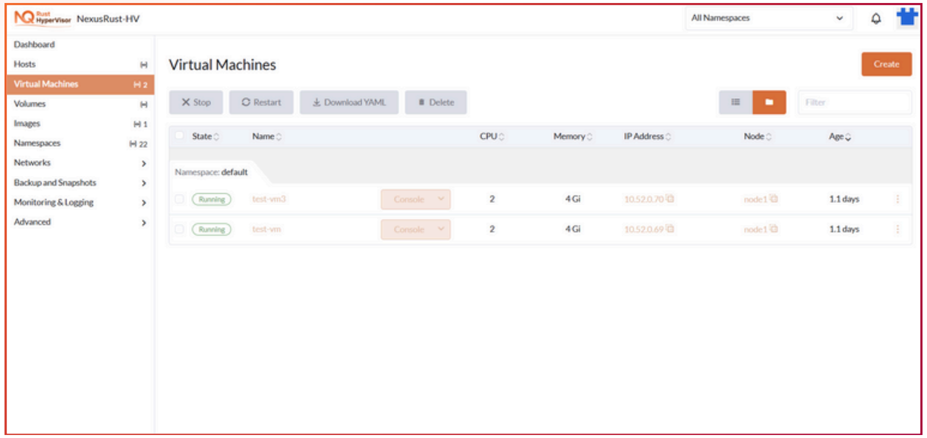
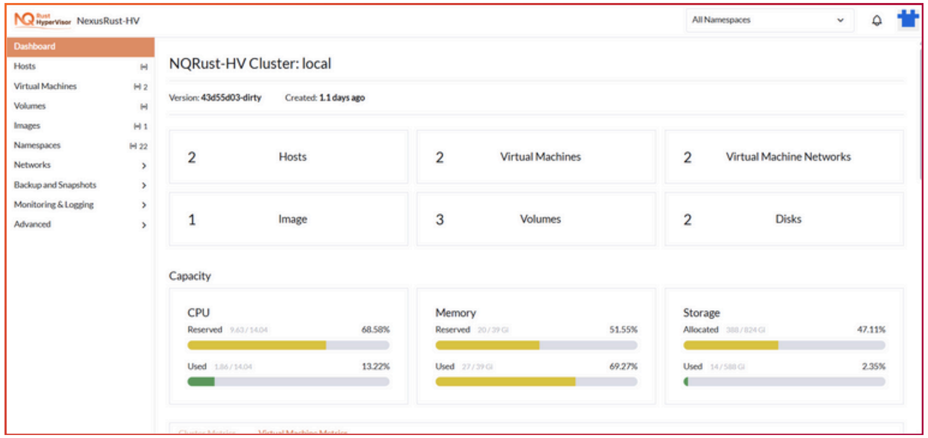
NQRust-HV Enterprise Architecture v2.0
October 2025 | Memory-Safe Virtualization | Nexus Quantum Tech

Nexus Quantum

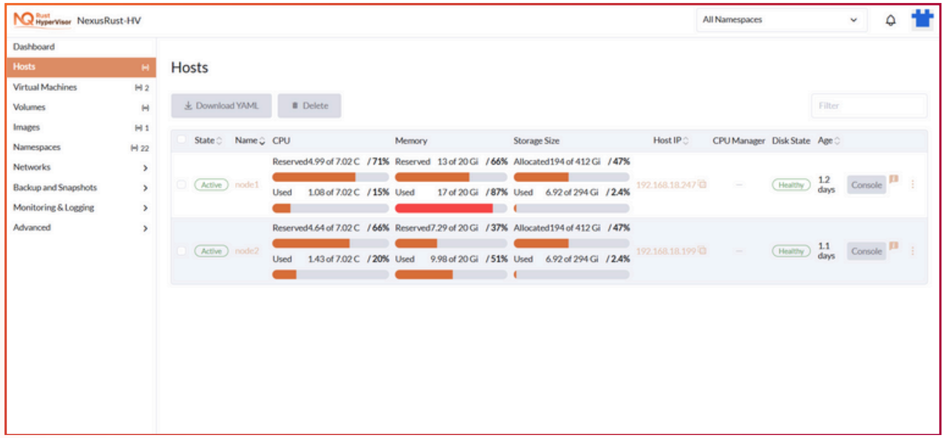
Product Overview





This page shows a simple overview of all virtual machines running on the NexusRust HyperVisor platform. It lists each virtual machine's current status, name, assigned capacity, network address, and how long it has been running, making it easy to see which services are active at a glance. The action buttons at the top allow users to create new virtual machines or quickly stop, restart, or remove existing ones, while the Console button provides direct access when needed.

This dashboard gives a simple, high-level overview of the NexusRust HyperVisor, showing the current status, scale, and available capacity of the system. It summarizes key resources such as servers, virtual machines, networks, and storage, while the side menu allows easy access to management, monitoring, and backup features.



This page shows a clear overview of all physical servers ("hosts") that power the NexusRust HyperVisor system. For each host, it displays its current status, name, and age, along with easy-to-read bars that show how much processing power, memory, and storage are available versus in use. The host IP and health indicator help users quickly confirm connectivity and stability, while action buttons like Console, Download, and Delete allow simple management when needed.